

Unit 3 – Collecting Data

Notes 2 – Potential Problems with Sampling

Sources of Bias

Samples are _____ if they are systematically not representative of the desired population.

- Bias occurs when certain responses are systematically favored over others.

_____ : When a sample is comprised entirely of volunteers or people who choose to participate, the sample will typically not be representative of the population (voluntary response bias).

- *Example:* A radio talk show hosts asks listeners to call in with their opinions of making wearing masks in public space mandatory.

_____ : Occurs when some groups in the population are left out of the process of choosing a sample.

- *Example:* Because they are generally fearful of government intrusion, many immigrants from Latin America did not return their census questionnaire during the 1990 census.

_____ : Occurs when an individual chosen for a sample can't be contact or refuses to respond. Non-response is a big problem in mail surveys.

- *Example:* Our administration sends out 100 survey questions to a sample of parents in order to gage their attitudes towards returning to school in 2020. Only 23 respond.

Response Bias - _____

_____ : people give untruthful answers for several reasons:

1) *Sensitive Questions:* How often do you drink alcohol?

2) *Socially Acceptable Answers:* Do you use corporal punishment with your children?

3) *Interview Bias:* One year after the Detroit race riots of 1967, interviewers asked a sample of black residents in Detroit if they felt they could trust most white people, some white people, or none at all. When the interviewer was white, 35% answered "most"; when the interviewer was black, 7% answered "most".

_____ : people will give silly answers just so that they won't appear like they know nothing about the subject.

Example: In a study educators were asked how they would rank Princeton's undergraduate business program. In every case, it was rated among the top 10 departments in the country, even though Princeton doesn't offer an undergraduate business major.

_____ : giving a wrong answer simply because respondent doesn't remember the correct answer.

Example: Students were asked to report their grade point averages. Researchers then determined the actual GPA's. Over 17% of the students reported a GPA that was .4 or more above their actual average, and about 2% reported a GPA more than .4 below their actual GPA. (most inflated their GPA's!)

_____ : When a survey is taken can have an impact on the answers.

Example: in January, the National Football League reported a poll that revealed football as the nation's favorite sport (this is at the time of the Super Bowl)

_____ : Subtle differences in phrasing make large differences in the results.

Example:

- a) Should the president have the line-item veto to eliminate waste? 97% said "yes"
- b) Should the president have the line-item veto? 57% said "yes"

When drawing a sample, two types of errors may occur:

Sampling Error: _____

Example: Place 50 red and 50 green balls in a bag. Mix the balls thoroughly and randomly sample 30 balls. In your sample you find that 12 balls are red and 18 are green. Your sample result (12:18 = 2:3) is different than the true population ratio of 50:50 which is 1 to 1. This difference is due to sampling error. Virtually any experiment involving a sample will have sampling error. We can minimize sampling error through various statistical techniques; the most obvious is to _____.

Non-sampling error: _____

Example: In order to gage student opinion on a new grading policy, an administrator stands outside the library during common time and asks a sample of 50 students if they agree with the new policy. The administrator finds that 25 out of the 50 students sampled agree with the new policy. When the entire student body is asked their opinion, however, the results were 30% in favor 70% against. The difference between the sample percentage (25/50 = 50%) and the true population percentage is due to non-sampling error, because the sample was collected in such a way that a lot of bias was involved (convenience sampling).